

Solution Requirements

Date	26 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can access the OptiCrop application to get crop recommendations.	High
2	Input Validation	The system validates N, P, K, temperature, humidity, pH, and rainfall values entered by the user.	High
3	External Interfaces	The system uses the trained machine learning model (model.pkl) and agricultural dataset.	High
4	Crop Prediction	The system predicts the most suitable crop based on the input values.	High
5	Result Display	The predicted crop is displayed clearly to the user.	High
6	Business Rules	Recommendations are generated only from valid agricultural input values.	Medium
7	Compliance	The system follows standard software development and data handling practices.	Medium
8	Other	The system can be updated with new datasets and improved ML models in the future.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should provide crop recommendations quickly.	Prediction within 2–3 seconds
2	Scalability	The application should support more users and larger datasets.	Supports multiple users
3	Security & Privacy	User input should be handled securely.	Secure processing of input data
4	Reliability & Availability	The application should work without frequent failures.	99% uptime during usage
5	Usability & Accessibility	The interface should be simple and easy to use.	User-friendly and responsive design
6	Other	The system should be easy to maintain and update.	Easy model replacement and maintenance